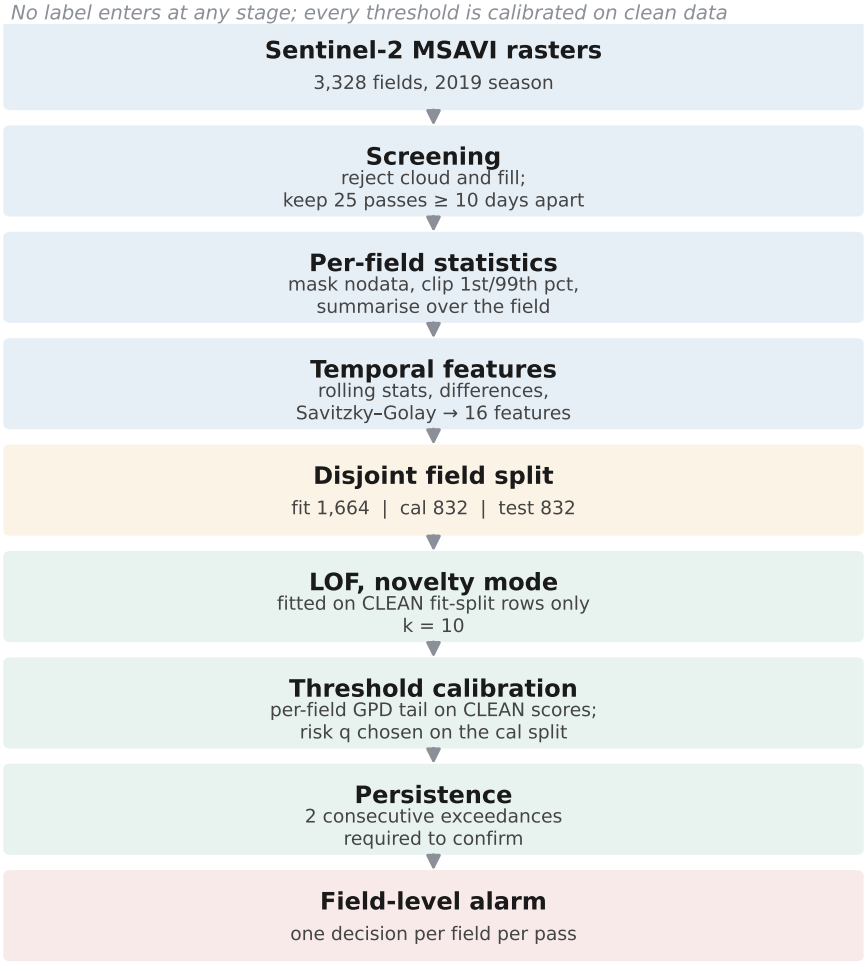Processing pipeline